ABSTRACT

The web application development internship provided me with hands-on experience in creating, designing, and deploying web applications using technologies such as HTML, CSS, JavaScript, and PHP. During this internship, I collaborated with experienced developers and designers, which helped me develop essential skills in building dynamic and responsive web applications. I became proficient in utilizing industry-standard tools and frameworks, focusing on front-end design using HTML and CSS, as well as back-end development using PHP and JavaScript. This allowed me to create user-friendly, efficient, and scalable web applications.

A significant aspect of my role involved writing clean, maintainable code, while ensuring best practices in web development were followed. I contributed to debugging and troubleshooting efforts, ensuring that the web applications performed smoothly and efficiently. My tasks included resolving performance issues, improving UI/UX elements, and addressing bugs. Additionally, I had the opportunity to integrate various third-party services, such as payment gateways, user authentication systems, and APIs, to enhance the overall functionality of the applications.

Testing was a key responsibility during my internship. I performed unit testing, UI testing, and cross-browser testing to ensure that the applications functioned well across different platforms and browsers. I worked closely with the design team to optimize the user interface and experience, ensuring that the applications were visually appealing, intuitive, and easy to navigate.

I also contributed to the deployment of web applications, gaining experience in hosting and managing applications on servers and cloud platforms, while ensuring they met security and performance standards. This experience helped me understand the complete web application development lifecycle, from the initial concept and design phase to deployment and ongoing updates. Overall, this internship greatly enhanced my technical expertise and provided me with a deeper understanding of the business, design, and user experience aspects of web application development.

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CHAPTER 1
INTRODUCTION TO WEB APPLICATION DEVELOPMENT

Web application development is the process of designing, building, and deploying applications that run in web browsers. It involves the integration of both frontend and backend technologies to create dynamic, responsive, and user-friendly websites and web-based applications. The advent of web technologies such as HTML, CSS, JavaScript, and PHP has revolutionized how applications are built, enabling developers to create sophisticated and interactive user experiences.

With the increasing reliance on the internet and web-based services, web application development has become a crucial aspect of business, education, entertainment, and communication. From simple static websites to complex enterprise applications, web development plays a key role in digital transformation. The need for scalable and secure web applications has also driven the evolution of modern web development practices and technologies, including the use of frameworks like Bootstrap for responsive design and PHP for backend integration.

Web applications can be accessed from any device with a browser and an internet connection, making them inherently flexible and platform-independent. Unlike traditional desktop applications, which require installation on each device, web applications can be deployed on servers and accessed remotely, offering the advantage of seamless updates and maintenance.

In this section, we will explore the foundational technologies that form the backbone of web application development, including frontend technologies such as HTML, CSS, and JavaScript, and backend technologies like PHP and MySQL. We'll also touch on frameworks such as Bootstrap, which aid in creating responsive and mobile-friendly web designs.

1.1 Key Technologies in Web Application Development:

The foundation of modern web development lies in the combination of frontend and backend technologies. Each plays a specific role in delivering a seamless web experience for users.

Frontend Development: Frontend development involves the creation of the visual components of a web application that users interact with. HTML (HyperText Markup Language) is the backbone of web content structure, providing the layout and elements of a page. CSS (Cascading Style Sheets) is used for styling, enabling developers to define the look and feel of web pages, including fonts, colors, and layouts. JavaScript adds interactivity to web pages, allowing for dynamic content and user-driven events.

HTML: HTML is the standard language used to structure content on the web. It defines the structure of a webpage using a series of elements such as headings, paragraphs, lists, and links. Each element represents a part of the web page and is displayed in the browser.

CSS: CSS is used to control the layout and design of web pages. With CSS, developers can style elements defined by HTML to make the page visually appealing. Techniques like flexbox and grid help in building complex layouts that adjust dynamically to screen sizes, a key aspect of responsive design.

JavaScript: JavaScript is a powerful scripting language that enables dynamic interactions on web pages. It can manipulate the HTML and CSS elements in realtime, allowing for actions such as form validation, dynamic content loading, and interactive elements like sliders and buttons.

Backend Development: The backend of a web application is responsible for handling data, processing requests, and ensuring the application runs smoothly. PHP is one of the most widely used server-side scripting languages for web development. It interacts with databases to store and retrieve data, handles authentication, and processes business logic.

PHP: PHP is a server-side language that is embedded within HTML to dynamically generate content for a web page. It is often used to handle form submissions, user authentication, and interact with databases. PHP integrates seamlessly with MySQL or other relational databases to store and retrieve information.

MySQL: MySQL is a widely used database management system for storing data in web applications. It stores user information, transaction data, and other applicationspecific data. PHP interacts with MySQL to create, read, update, and delete data in the database (CRUD operations).

1.2 Importance of Responsive Design and User Experience:

The success of a web application is closely tied to its design and usability. With a variety of devices used to access web applications, it is essential that web applications are designed to be responsive. Responsive web design ensures that an application looks and functions well across all devices, including desktops, tablets, and smartphones.

Responsive Web Design: The rise of mobile internet users has made responsive design a key priority for developers. Tools like **Bootstrap** offer a framework that automatically adjusts web page layouts to fit various screen sizes, providing a seamless user experience across all devices. Using **media queries** in CSS allows developers to apply different styles for different screen sizes, ensuring that elements are properly aligned and + on all devices.

Bootstrap: Bootstrap is a front-end framework that helps developers build responsive websites quickly and easily. It comes with predefined CSS and JavaScript components for designing navigation bars, buttons, grids, and more. It is widely used due to its flexibility, speed, and ease of integration.

User Experience (UX): UX design is crucial in web development as it focuses on the overall experience of the user while interacting with a website or application. A good UX ensures that the web application is intuitive, easy to navigate, and meets the needs of its users. This involves focusing on aspects like navigation, loading times, accessibility, and visual hierarchy. By improving UX, developers ensure higher user satisfaction, engagement, and retention.

UI/UX Design Collaboration: Collaboration between developers and designers is essential in creating web applications that not only function well but are also visually appealing. UI (User Interface) design focuses on the layout, appearance, and interactivity of a web application, while UX (User Experience) design focuses on the overall user journey. Combining both disciplines results in a more polished and user-friendly application.

User Testing: User testing is a critical part of UX design. It involves gathering feedback from users to identify pain points and areas for improvement.Usability testing helps developers understand how usersinteract with the web application and provides valuable insights into how to improve the design and functionality.

CHAPTER 2

PROJECT OVERVIEW AND OBJECTIVES (NEWSBUNCH EXAMPLE)

2.1 Project Overview:

NewsBunch is a dynamic web application designed to provide users with the latest news across various categories such as sports, politics, movies, and more. The application allows users to view, search, comment, and engage with the news content based on their interests. With multiple sections such as top stories, most viewed news, and separate local, national, and international news categories, NewsBunch offers a tailored experience for users who want to stay informed about the world.

The app's primary features include search functionality, where users can search news articles by keywords or categories; user logins, allowing users to create accounts and save their preferences or liked articles; and real-time updates, ensuring users always see the latest news.

Developed with PHP for the backend, MySQL for data storage, and HTML, CSS, and JavaScript for the frontend, NewsBunch ensures that the web application is both user-friendly and responsive across various devices. It was designed to be scalable, optimized for performance, and secure to handle potential high traffic and user interactions.

2.2 Deliver a Fully Functional News Web Application:

The main objective of this project was to build a fully functional news website that would provide users with timely and accurate news updates. It was crucial to ensure that users could easily navigate through categories, search for news, and access articles with a smooth and responsive interface.

- Example: Implementing dynamic categories such as top stories, sports, and movies, with each category having its own set of articles displayed based on the most recent or most viewed content.

2.3 Ensure Easy Deployment and Hosting:

The objective was to deploy NewsBunch to a live environment where users could access the application. This involved selecting the right hosting solution, configuring the server environment, ensuring security, and implementing auto-scaling to handle traffic spikes effectively.

- Example: Deploying NewsBunch on AWS EC2 instances with auto-scaling enabled to ensure the application could handle high traffic, especially during peak news events.

2.4 Implement User Authentication and Personalized Features:

A key feature of NewsBunch was the ability to allow users to create accounts, log in, and access personalized content such as liked articles, saved news for later reading, and comments on articles. The objective was to allow users to create a personalized experience while maintaining secure authentication.

Developed with PHP for the backend, MySQL for data storage, and HTML, CSS, and JavaScript for the frontend, NewsBunch ensures that the web application is both user-friendly and responsive across various devices. It was designed to be scalable, optimized for performance, and secure to handle potential high traffic and user interactions**.**

web application is closely tied to its design and usability. With a variety of devices used to access web applications, it is essential that web applications are designed to be responsive. Responsive web design ensures that an application looks and functions well across all devices, including desktops, tablets, and smartphones

- Example: Using password hashing and session management to ensure secure user login and personalized content access.

CHAPTER-3

LEARNING AND IMPLEMENTING HTML, CSS, AND JAVASCRIPT

Web development is divided into two main parts: **frontend** and **backend** development. The **frontend** of a web application is the part that users directly interact with, and it plays a crucial role in ensuring that users have a positive experience while using the application. The foundation of frontend development lies in three key technologies: **HTML (HyperText Markup Language)**, **CSS (Cascading Style Sheets)**, and **JavaScript.**

3.1 HTML (HyperText Markup Language):

HTML is the fundamental language used to create the structure and content of a webpage. It consists of a series of elements, often referred to as **tags**, that define the different sections and components of a web page. These tags include headings, paragraphs, links, images, and more, which are displayed by web browsers in a structured format.

• **Structuring Content with HTML**: The primary purpose of HTML is to structure the content of a website. For instance, <h1> to <h6> tags are used to define headings, while <p> is used for paragraphs, <a> for links, and <img> for images. These tags help organize content and make it easy for browsers to display the page.

o **Example:** html <h1>Welcome to My Website</h1>

<p>This is a simple paragraph explaining the content of the page.</p>

Visit Example

• **Semantic HTML**: In addition to basic HTML tags, it is important to use **semantic HTML**, which involves using tags that convey the meaning of the content they enclose. Tags such as <header>, <footer>, <article>, and <section> help improve accessibility, SEO, and overall code readability. For example, the <header> tag is used to define the header section of the page, and the <footer> tag represents the footer section.

o **Benefits of Semantic HTML**: - **Improved Accessibility**: Screen readers can interpret the page structure more easily, making the website more accessible to users with disabilities. - **Better SEO**:

Search engines understand the structure and hierarchy of the content better, which can lead to improved search rankings.

o **Example:** html <header> <h1>My Web Application</h1>

```
<nav>
<ul>
<li><a href="#">Home</a></li>
<li><a href="#">About</a></li>
 </ul>
 </nav>
</header>
```

• **HTML Forms and Input Elements**: Forms are an essential part of web applications, allowing users to submit data. HTML provides a variety of input elements like <input>, <textarea>, <select>, and <button> for gathering user input.

o **Example:** <form action="/submit" method="POST">

```
<label for="name">Name:</label>
<input type="text" id="name" name="name">
<button type="submit">Submit</button>
 </form>
```

3.2 CSS (Cascading Style Sheets):

CSS is used to style and design the visual aspects of a webpage. While HTML provides the structure, CSS controls the layout, colors, fonts, and other visual elements. CSS allows developers to separate content from presentation, making it easier to maintain and update the design without changing the underlying HTML structure.

• **Styling Web Pages with CSS**: CSS is applied to HTML elements to change their appearance. This includes altering properties like color, background, font size, margins, and padding.
Developers can use CSS selectors to target specific elements or groups of elements on a webpage.

o **Example:** ```css body

{

font-family: Arial, sans-serif; background-color:

#f4f4f4;

} h1 {

color: #333;

font-size: 36px;

}

```

• **Responsive Design**: In today's world, users access web applications from a variety of devices, including smartphones, tablets, and desktops. Therefore, it is essential to create websites that adjust to different screen sizes and resolutions. **Responsive web design** (RWD) ensures that web applications are optimized for various devices by using CSS media queries. These media queries allow developers to apply different styles based on the screen size, resolution, or device type.

o **Example (CSS Media Query)**: css

```
@media (max-width: 768px) { body
    {
        font-size: 14px;
    }
  }
```

• **Benefits of Responsive Design**: - **Enhanced User Experience**: Users can enjoy the seamless browsing experience regardless of the device they use. - **Mobile-Friendly Websites**: With the increasing use of mobile devices, responsive design ensures that users can interact with websites effectively on smartphones and tablets.

• **CSS Flexbox and Grid**: CSS Flexbox and Grid are powerful layout tools that help developers create complex and flexible web page layouts. These tools allow elements to be aligned, spaced, and resized dynamically, ensuring the layout adapts to different screen sizes.
```

Example (CSS Flexbox): ```css

.container { display: flex;

justify-content: space-between;

}

3.3 JavaScript:

JavaScript is a versatile and powerful programming language that adds interactivity to web pages. Unlike HTML and CSS, which focus on structure and styling, JavaScript handles the behavior and functionality of a web application. It allows developers to create dynamic web pages that respond to user actions, such as clicking a button or submitting a form.

- **Interactivity with JavaScript**: JavaScript enables developers to create interactive elements like forms, modals, dropdown menus, and image sliders. It responds to user actions like clicks, mouse movements, key presses, and more.

- **DOM Manipulation**: The **Document Object Model (DOM)** represents the structure of an HTML document. JavaScript can be used to manipulate the DOM, allowing developers to modify the content, attributes, and style of elements on a webpage in real-time.

- **Benefits of DOM Manipulation**: - **Real-Time Updates**: Changes to the DOM can be made instantly without reloading the page, resulting in a smoother user experience. - **Dynamic Content**: JavaScript allows for dynamic content loading, such as fetching data from APIs and updating the page without a full reload.

- **Form Validation and Error Handling**: JavaScript is often used for validating form input before submitting it to the server. This helps ensure that users provide correct and complete data, preventing errors on the server side.

- **AJAX (Asynchronous JavaScript and XML)**: AJAX is a technique that allows web pages to retrieve data from the server asynchronously, without refreshing the entire page. This results in a faster, more dynamic user experience. AJAX is often used to load new content or submit forms without a full page reload.

CHAPTER 4

INTRODUCTION TO BOOTSTRAP FRAMEWORK

Bootstrap is a popular front-end framework that simplifies the process of building responsive, mobile-first websites. It provides a collection of pre-designed CSS and JavaScript components that allow developers to quickly create well-structured, aesthetically pleasing, and functional web applications. Whether you're building a simple landing page or a complex web application, Bootstrap helps you streamline the development process with its grid system, components, and utilities.

4.1 Responsive Grid System:

One of the core features of Bootstrap is its **responsive grid system**. This system allows you to create fluid layouts that automatically adjust to the screen size of the device being used. The grid is based on 12 columns, and you can adjust the layout by assigning different numbers of columns to different screen sizes (e.g., large, medium, small).

Example of Bootstrap Grid System:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Bootstrap Grid System</title>
  <link href=https://stackpath.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.min.css
rel="stylesheet">
</head>
<body>
  <div class="container">
    <div class="row">
      <div class="col-lg-4 col-md-6 col-12">
```

```html
        <div class="p-3 border bg-light">Column 1</div>
      </div>
      <div class="col-lg-4 col-md-6 col-12">
<div class="p-3 border bg-light">Column 2</div>
      </div>
      <div class="col-lg-4 col-md-12 col-12">
        <div class="p-3 border bg-light">Column 3</div>
      </div>
    </div>
  </div>
  <script src="https://code.jquery.com/jquery-3.5.1.slim.min.js"></script>
  <script
src="https://cdn.jsdelivr.net/npm/@popperjs/core@2.5.2/dist/umd/popper.min.js"></script>
  <script
src="https://stackpath.bootstrapcdn.com/bootstrap/4.5.2/js/bootstrap.min.js"></script>
</body>
</html>
```

In this example:

- The **col-lg-4** class means that on large screens (LG), the column will take up 4/12 of the container (1/3 of the screen).
- The **col-md-6** class means that on medium screens (MD), the column will take up 6/12 (half the screen).
- The **col-12** class ensures that on smaller screens (XS), the column will take up the full screen width.
- This grid system allows the layout to adjust based on the screen size, making it responsive.

4.2 Pre-built Components:

Bootstrap comes with a range of pre-designed components like buttons, navigation bars, cards, and modals, which can be easily customized to fit your design. These components help developers quickly implement commonly used UI elements without having to write a lot of CSS.

1. **Example of a Bootstrap Button**:

```
<!DOCTYPE html>

<html lang="en">

<head>

 <meta charset="UTF-8">

   <meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Bootstrap Button</title>

   <link        href="https://stackpath.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.min.css"
rel="stylesheet">

</head>

<body>

   <div class="container mt-5">

      <button type="button" class="btn btn-primary">Primary Button</button>

      <button type="button" class="btn btn-secondary">Secondary Button</button>

      <button   type="button"   class="btn   btn-success">Success   Button</button>

</div>

   <script src="https://code.jquery.com/jquery-3.5.1.slim.min.js"></script>
```

```
<script
src="https://cdn.jsdelivr.net/npm/@popperjs/core@2.5.2/dist/umd/popper.min.js"></script>

<script
src="https://stackpath.bootstrapcdn.com/bootstrap/4.5.2/js/bootstrap.min.js"></script>
</body>
</html>
```

In this example:

- The **btn** class is used to style a button.
- Additional classes like **btn-primary**, **btn-secondary**, and **btn-success** determine the button’s color scheme.
- This makes it easy to create consistent button designs with minimal effort.

4.3 Navigation Bar:

Bootstrap provides an easy-to-use navigation bar component that helps in building responsive navigation menus. The navbar adapts itself to different screen sizes, and it can collapse into a mobile-friendly version with a hamburger menu on smaller screens.

Example of a Bootstrap Navigation Bar:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0"
<title>Bootstrap Navbar</title>
  <link href=https://stackpath.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.min.css
rel="stylesheet">
</head>
```

```
<body>
    <nav class="navbar navbar-expand-lg navbar-light bg-light">
        <a class="navbar-brand" href="#">Navbar</a>
        <button class="navbar-toggler" type="button" data-toggle="collapse"
datatarget="#navbarNav" aria-controls="navbarNav" aria-expanded="false" aria-label="Toggle
navigation">
        </button>
        <div class="collapse navbar-collapse" id="navbarNav">
            <ul class="navbar-nav">
                <li class="nav-item active">
                    <a class="nav-link" href="#">Home <span class="sr-only">(current)</span></a>
                </li>
                <li class="nav-item">
                    <a class="nav-link" href="#">Features</a>
                </li>
                <li class="nav-item">
                    <a class="nav-link" href="#">Pricing</a>
                </li>
                <li class="nav-item">
                    <a class="nav-link disabled" href="#" tabindex="-1" disabled="true">Disabled</a>
                 </li>
            </ul>
        </div>
          </body>
</html>
```

CHAPTER 5

BACKEND DEVELOPMENT WITH PHP (NEWSBUNCH EXAMPLE)

Backend development in the **NewsBunch** web application is powered by **PHP**, a widelyused server-side scripting language. PHP handles the dynamic generation of news content, user authentication, database interactions, and API integration, which are crucial for any content-heavy web application like NewsBunch.feature uses user data, such as liked articles, commented content, and saved stories, to curate a feed that is tailored to each individual's interests. Personalized recommendations help improve user retention, as users are more likely to stay engaged with an app that consistently delivers content that aligns with their preferences.

• **Database Interaction**: PHP works with MySQL to manage articles, categories (sports, politics, movies), and user data. For example, fetching the latest news articles or handling user logins requires PHP to connect to the database, query the necessary data, and serve it to the frontend.

Example Code (Fetching Latest News from the Database):

```
<?php
// Database connection
$connection = new mysqli('localhost', 'username', 'password', 'newsbunch');
// Check connection if ($connection->connect_error) {
die("Connection failed: " . $connection->connect_error);
}
// Fetch latest news for top stories
$query = "SELECT * FROM articles ORDER BY publish_date DESC LIMIT 5";
$result = $connection->query($query);
// Display top stories while ($row =
$result->fetch_assoc()) {     echo "<div
class='news-article'>";     echo "<h3>" .
```

```
$row['title'] . "</h3>";     echo "<p>" .
$row['summary'] . "</p>";     echo
"</div>";
}
$connection->close();
?>
```

In this example, PHP connects to a MySQL database and fetches the latest 5 articles to display as **top stories**.

User Authentication: User login and registration functionalities are built using PHP. Users can log in to view personalized content or interact with articles.

Example Code (User Login Logic):

```
<?php
session_start();
if ($_SERVER["REQUEST_METHOD"] == "POST") {
   $username = $_POST['username'];
   $password = $_POST['password'];
if ($result->num_rows > 0) {
      $_SESSION['user']  =  $username;    //  Store  user  session
echo "Login successful!";
   } else {          echo "Invalid username or
password!";
   }
   $connection->close();
}
?>
```

CHAPTER 6

FRONTEND AND BACKEND INTEGRATION (NEWSBUNCH EXAMPLE)

Integrating the frontend with the backend is a crucial step in building any web application. In NewsBunch, frontend technologies like HTML, CSS, and JavaScript communicate with the backend (PHP) to dynamically load news articles, handle user input (like search), and manage session states.

• **AJAX for Dynamic News Updates**: **AJAX** allows the page to load news articles without refreshing. It provides a seamless user experience where users can interact with the website without waiting for the whole page to reload.

Example of AJAX for Fetching Latest News:

```
 function fetchNews(category) {
var xhr = new XMLHttpRequest();
xhr.open('GET', 'fetch_news.php?category=' + category, true);
xhr.onload = function() {
if (xhr.status === 200) {
 document.getElementById('news-container').innerHTML = xhr.responseText;
       }
 };
xhr.send();
     }
```

In this example, an AJAX request is sent to the PHP backend to fetch news articles from a particular category (like **sports** or **politics**) and update the page without reloading.

CHAPTER 7

PERFORMANCE OPTIMIZATION IN WEB APPLICATIONS (NEWSBUNCH EXAMPLE)

Optimizing the performance of NewsBunch ensures that the application loads quickly and provides a smooth user experience, especially given that it features categories like top stories, most viewed, and multiple news types.

7.1 Database Optimization:

As **NewsBunch** grows, the number of articles in the database will increase. To improve the performance of database queries, it's important to index frequently accessed fields such as **article titles**, **categories**, and **publish dates**.

SQL Injection is one of the most common and dangerous security vulnerabilities in web applications. It occurs when an attacker manipulates SQL queries by injecting malicious SQL code through user inputs, potentially allowing unauthorized access to the database or modification of data. In NewsBunch, we prevent SQL injection by using prepared statements and parameterized queries, ensuring that user input is safely handled**.**

Storing passwords in plain text is a significant security risk. If the password database is compromised, the attackers can easily access all user accounts. NewsBunch implements secure password handling by using password hashing and password verification methods provided by PHP**.**

Example Code (Optimizing Database Queries):

```
// Indexing the 'category' and 'publish_date' columns for faster queries

$query = "SELECT * FROM articles WHERE category = ? ORDER BY publish_date DESC
LIMIT 5";

$stmt = $connection->prepare($query);

$stmt->bind_param("s", $category);

$stmt->execute();
```

This query is optimized by using indexes on the category and publish_date columns, allowing it to fetch relevant articles more efficiently.

CHAPTER 8

WEB APPLICATION SECURITY BEST PRACTICES (NEWSBUNCH EXAMPLE)

Security is a top priority for any web application, especially one like **NewsBunch**, where sensitive user data (including usernames, passwords, and other personal information) needs to be protected, and the integrity of the news content must be ensured. As news applications are a popular target for malicious activities due to their content and potential traffic, implementing security best practices is critical. Below are the key security practices implemented in **NewsBunch** to safeguard both user and application data.

8.1 SQL Injection Prevention:

SQL Injection is one of the most common and dangerous security vulnerabilities in web applications. It occurs when an attacker manipulates SQL queries by injecting malicious SQL code through user inputs, potentially allowing unauthorized access to the database or modification of data. In NewsBunch, we prevent SQL injection by using prepared statements and parameterized queries, ensuring that user input is safely handled**.**

• **Prepared Statements in PHP:** Prepared statements are a powerful defense against SQL injection. By using prepared statements, we separate SQL code from user input, ensuring that the user input is treated as data rather than executable code.

Example Code (Prepared Statements in PHP):

```
<?php

// Database connection

$connection = new mysqli('localhost', 'username', 'password', 'newsbunch');

// Check connection if ($connection->connect_error) {

die("Connection failed: " . $connection->connect_error);

}

// Fetch articles by category using prepared statements to avoid SQL injection

$query = "SELECT * FROM articles WHERE category = ? ORDER BY publish_date DESC
LIMIT 10";
```

```
$stmt = $connection->prepare($query);

$stmt->bind_param("s", $category); // Binding the user-provided category safely

$stmt->execute();

$result = $stmt->get_result(); // Display

the articles while ($row = $result-

>fetch_assoc()) {     echo "<div

class='news-article'>";     echo "<h3>" .

$row['title'] . "</h3>";     echo "<p>" .

$row['summary'] . "</p>";     echo

"</div>";

}

$stmt->close();

$connection->close();

?>
```

In this example, the prepared statement safely binds the user input ($category) as a parameter, making it impossible for an attacker to inject malicious SQL code. By using bind_param(), we ensure that the input is treated as a string and is securely handled by the database.

8.2 Password Hashing and Verification:

Storing passwords in plain text is a significant security risk. If the password database is compromised, the attackers can easily access all user accounts. NewsBunch implements secure password handling by using password hashing and password verification methods provided by PHP**.**

Hashing Passwords with password_hash():

PHP's password_hash() function is used to hash passwords before storing them in the database. This ensures that even if an attacker gains access to the database, they cannot retrieve the original passwords. The hash is one-way, meaning it cannot be reversed to reveal the original password**.**

Example Code (Hashing Passwords):

```php
<?php
// Hashing the password using PHP's password_hash() function
$password = "userPassword123"; // User-provided password
$hashedPassword = password_hash($password, PASSWORD_DEFAULT);


// Store $hashedPassword in the database (never store plain-text passwords!)
?>
```

In this example:

password_hash() uses the **bcrypt** algorithm by default to securely hash the password.

The hash is stored in the database instead of the plain-text password.

Verifying Passwords with password_verify(): During login, NewsBunch compares the provided password with the stored hash using password_verify(). This function checks whether the entered password, when hashed, matches the stored hash.

Example Code (Password Verification):

```php
<?php
// Example of verifying user-provided password against the stored hash
$inputPassword = "userPassword123"; // User's entered password
$storedHash = '$2y$10$C.4...'; // Retrieved hash from the database
if (password_verify($inputPassword, $storedHash)) {     echo
"Password is correct!";
}  else  {         echo  "Invalid
password!";
} ?>
```

CHAPTER 9

TESTING AND DEBUGGING (NEWSBUNCH EXAMPLE)

Testing and debugging are critical steps in the development process for ensuring that the NewsBunch web application functions correctly and is free from errors. Rigorous testing ensures that features such as user login, article fetching, search functionality, and category filtering work as intended. Debugging helps identify and fix issues that may arise during the development and deployment phases. Below, we will explore different aspects of testing and debugging for NewsBunch.

9.1 Unit Testing PHP Functions (NewsBunch Example):

Unit testing involves testing individual components (functions, methods) of the application to ensure they behave as expected. For **NewsBunch**, unit testing can be used to test backend logic such as fetching news articles, user authentication, and search functionality.

• **PHP Unit for Unit Testing:**

PHPUnit is a popular testing framework for PHP. In **NewsBunch**, we use PHPUnit to write unit tests for various PHP functions such as fetching articles by category, handling search queries, and verifying user login logic.

Example Code (Unit Test for Fetching Articles by Category):

```
// testNewsFunctions.php

use PHPUnit\Framework\TestCase; class

NewsTest extends TestCase

{

public function testFetchArticlesByCategory()

   {

       // Simulate a database connection (mocking the database interaction)

 $mockDb = $this->createMock(mysqli::class);

       $mockDb->method('query')

           ->willReturn(true);
```

```
$newsApp = new NewsApp($mockDb);

        // Call the function to test

        $result = $newsApp->fetchArticlesByCategory('Sports');

 // Assert that the result is an array (i.e., articles returned)

        $this->assertIsArray($result);

    }

}
```

9.2 End-to-End Testing (NewsBunch Example):

End-to-end testing (E2E) tests the entire application flow from the user's perspective. It ensures that all parts of the application, from the frontend (HTML, CSS, JavaScript) to the backend (PHP, MySQL), work together seamlessly.

```
// e2eTest.js const { Builder, By, until } =

require('selenium-webdriver'); const driver = new

Builder().forBrowser('chrome').build(); async function

testLogin() {

    try {

    driver.get('http://localhost/newsbunch/login.php');

driver.findElement(By.name('username')).sendKeys('john_doe');

driver.findElement(By.name('password')).sendKeys('password123');

driver.findElement(By.name('submit')).click();

driver.wait(until.urlIs('http://localhost/newsbunch/dashboard.php'), 5000);

const pageTitle = await driver.getTitle();

console.log("Page title is: " + pageTitle);

    }

} testLogin();
```

CHAPTER 10
DEPLOYMENT AND HOSTING OF WEB APPLICATIONS (NEWSBUNCH EXAMPLE)

Once a web application like NewsBunch has been developed and thoroughly tested, the next critical phase is deploying it to a live environment. This is where the application becomes publicly accessible and users can begin interacting with it. Deployment and hosting involve configuring the server, uploading files, setting up databases, ensuring scalability, and implementing security measures. The process begins with selecting the appropriate hosting solution, followed by configuring the server environment, ensuring the application's functionality, optimizing for performance, and ultimately ensuring the application is secure and capable of handling expected traffic.

10.1 Selecting the Right Hosting Solution :

The first challenge when deploying **NewsBunch** was selecting the appropriate hosting solution that could handle both the traffic volume and the dynamic nature of the application. Initially, I considered **shared hosting** as a low-cost solution. However, as the application involved dynamic content generation (such as fetching articles, displaying top stories, and handling user logins), shared hosting proved to be insufficient due to limitations in terms of server resources, performance, and control over the environment.

- **Solution**: I opted for **cloud hosting** on **Amazon Web Services (AWS)**, specifically using **EC2** (Elastic Compute Cloud) instances. Cloud hosting provides scalability and flexibility, allowing the infrastructure to grow with the application. Additionally, AWS offers features such as **auto-scaling** and **load balancing**, which are necessary for handling high traffic during peak news events or breaking news stories.
- **Example**:

```
// Example of launching an EC2 instance with AWS CLI aws ec2 run-instances --image-id
ami-0c55b159cbfafe1f0 --count 1 --instance-type t2.micro
--key-name MyKeyPair
```

10.2 Setting Up the Server Environment

Once the hosting solution was chosen, the next step was to set up the **server environment**. For **NewsBunch**, the application was built using **PHP** for the backend, **MySQL** for the database, and **HTML**, **CSS**, and **JavaScript** for the frontend. To ensure that **NewsBunch** ran efficiently, I needed to configure a **LAMP stack** (Linux, Apache, MySQL, PHP) on the cloud server.

- **Solution**: On the **AWS EC2** instance, I installed **Apache** as the web server, **PHP** for the backend logic, and **MySQL** to handle the data storage. This environment was chosen because it is well-suited for PHP applications and offers ease of use and management for dynamic websites like **NewsBunch**. I also configured **SSL certificates** using **Let's Encrypt** to secure the communication between the user's browser and the server.
- **Example**:

```
// Installing Apache and PHP on Ubuntu (for AWS EC2 instance)
sudo apt-get update sudo apt-get install apache2 php libapache2-
mod-php mysql-server
```

10.3 Transferring Files to the Server

After the server environment was set up, I transferred the **NewsBunch** application files to the server. This included PHP scripts, HTML templates, CSS stylesheets, JavaScript files, and image assets.

- **Solution**: I used **SSH** (Secure Shell) to securely connect to the server and **SCP** (Secure Copy Protocol) to transfer files from the local development machine to the live server. For ongoing deployment and to manage changes in the codebase, I also integrated **Git** to pull the latest changes directly from the **GitHub** repository to the server.
- **Example**:

```
// Example of using SCP to transfer files scp -r /path/to/local/project
username@ec2-ip-address:/path/to/remote/directory
```

10.4 Database Migration

Since **NewsBunch** relies heavily on MySQL to store articles, user data, and comments, it was essential to migrate the local development database to the production server. The database had to be carefully transferred without losing any data.

- **Solution**: I exported the local MySQL database using **MySQL Dump**, transferred the dump file to the server, and imported it into the production MySQL instance. Additionally, I wrote **migration scripts** to handle any future schema changes to ensure that updates to the database structure could be easily applied across environments.
- **Example**:

```
// Exporting local database mysqldump -u root -p
newsbunch > newsbunch_backup.sql
// Importing to production mysql -u root -p newsbunch
< newsbunch_backup.sql
```

10.5 Configuring Web Server and URL Rewriting

One of the primary features of **NewsBunch** is the ability to display articles and categories with clean URLs, such as /sports, /politics, and /article/1234. Initially, the **URL rewriting** feature didn't work correctly in the production environment.

- **Solution**: I configured **Apache's .htaccess** file to enable **URL rewriting** and ensure that requests to dynamic pages like article details and category pages were routed correctly to the respective PHP scripts.
- **Example**:

```
# Enabling URL rewriting in Apache's .htaccess file
RewriteEngine On
RewriteRule ^article/([0-9]+)$ /article.php?id=$1 [L]
RewriteRule ^category/([a-zA-Z0-9-]+)$ /category.php?name=$1 [L]
```

CHAPTER 11

CHALLENGES FACED DURING THE INTERNSHIP (NEWSBUNCH EXAMPLE)

11.1 Handling Large Volumes of Data:

As **NewsBunch** contains a large number of articles, user comments, and media, it was essential to optimize the database queries for better performance. Initially, queries were slow due to the large dataset.

- **Solution**: Added **indexes** to frequently queried columns like category and publish_date and implemented **pagination** and **lazy loading** for fetching articles in batches rather than all at once.

  Example:

```
// Query with pagination to fetch articles

$page = isset($_GET['page']) ? (int)$_GET['page'] : 1;

$limit = 10;

$offset = ($page - 1) * $limit;

$query = "SELECT * FROM articles ORDER BY publish_date DESC LIMIT $limit OFFSET
$offset";
```

11.2 Cross-BrowserCompatibility:

The application's layout and components were displaying inconsistently across different browsers, especially on **Internet Explorer** and **Safari.**

- **Solution**: Used **Bootstrap** for responsive design and pre-built components, and implemented **CSS prefixes** and **media queries** for better compatibility.
- **Example**:

```
/* For Safari and older browsers */

.header {     display: -

webkit-flex;     display:

flex;
```

```
}
@media (max-width: 768px) {
  .header {         background-
color: #fff;
 }
}
```

11.3 Managing News Categories (Top Stories, Most Viewed, etc.)

Initially, the logic to dynamically fetch **top stories** or **most viewed** articles was difficult to manage, leading to inconsistent results.

- **Solution**: Refined the query logic to ensure articles were categorized correctly and used **AJAX** to dynamically load content for each category without refreshing the page.
- **Example**:

```
// AJAX request to fetch articles based on category function
fetchArticles(category) {
  $.ajax({        url:
'fetch_news.php',        data: {
category: category },  method:
'GET',        success:
function(data) {
          $('#news-container').html(data);
      }
  });
}
```

CHAPTER 12

CONCLUSION AND KEY TAKEAWAYS (NEWSBUNCH EXAMPLE)

The internship experience of developing the NewsBunch web application provided valuable hands-on exposure to real-world web development practices, from initial planning to deployment and scaling. It was an excellent opportunity to apply theoretical knowledge to practical situations, enhance technical skills, and face challenges that arise in the lifecycle of a production-grade web application.

Throughout the project, I had the chance to work with a variety of tools, technologies, and concepts. I gained in-depth experience with PHP, MySQL, JavaScript, and CSS, as well as frameworks like Bootstrap. These technologies enabled me to create a user-friendly, dynamic, and responsive web application that effectively displays and organizes news content in multiple categories. I also learned how to implement key features such as user authentication, search functionality, pagination, and dynamic content loading to improve the user experience.

One of the most significant aspects of the internship was learning about performance optimization. I had the chance to implement caching, lazy loading, and database indexing, which greatly improved the speed and responsiveness of the application, especially when dealing with large datasets. I also learned how to scale the application using cloud hosting and auto-scaling features, which ensured that the platform could handle an increase in traffic without compromising performance.

Security was another key area that I focused on during the internship. I applied best practices for user authentication (using password hashing and session management), SQL injection prevention (through prepared statements), and XSS and CSRF protection to safeguard the application from vulnerabilities. I also implemented SSL certificates to ensure secure data transmission, ensuring that user data and content remained protected.

Key Takeaways:

1. **Real-World Application Development**: Building **NewsBunch** provided practical experience in full-stack development, allowing me to work with backend (PHP, MySQL) and frontend (JavaScript, HTML, CSS) technologies while ensuring the application worked smoothly.

2. **Optimizing for Performance**: The importance of optimizing database queries, implementing caching mechanisms, and using performance-enhancing techniques like lazy loading and indexing became evident. These strategies drastically improved the application's speed and responsiveness.

3. **Security Best Practices**: Implementing security measures such as **password hashing**, **SQL injection prevention**, **XSS protection**, and **SSL encryption** was crucial in safeguarding user data and preventing malicious attacks.

4. **Scalability and Hosting**: Understanding how to scale an application and set it up on **cloud platforms** like **AWS** was a key learning experience. Configuring **auto-scaling**, **load balancing**, and ensuring the server could handle high traffic volumes are essential in maintaining a stable web application.

5. **Testing and Debugging**: Writing **unit tests**, using **debugging tools**, and performing rigorous testing were essential practices that ensured **NewsBunch** was functional, stable, and bug-free. This experience helped me appreciate the importance of testing and quality assurance in software development.

6. **Deployment and Maintenance**: Learning how to deploy the application, manage database mirations, configure the web server, and ensure the application remained secure and up-todate was an invaluable part of the internship.

7. **User-Centric Design**: Finally, ensuring the application was easy to navigate and responsive on multiple devices made me realize the importance of **user experience (UX)**. Every decision, from design to functionality, should always prioritize the user.

CONCLUSION:

In conclusion, my time developing NewsBunch was an enriching and invaluable experience. The challenges I faced and the lessons I learned have made me more confident in my ability to build secure, scalable, and performant web applications. The skills I gained during this internship will not only help me in future projects but also serve as a foundation for more advanced topics in web .